

Biosphere Dynamic Empower: Howard T. Odum's Contributions to Open Systems Thermodynamics

Mark T. Brown
Professor Emeritus
Environmental Engineering Science
University of Florida

Abstract:

Howard T. Odum's work in open systems thermodynamics (OST) revolutionized our understanding of energy flow within ecosystems and economies. Through concepts such as energy quality (emergy), the Maximum Power Principle, and hierarchical organization, he demonstrated how systems self-organize to maximize useful energy transformation. This paper reviews Odum's foundational contributions to OST—through emergy accounting, energy systems modeling, and systems ecology—and extends his vision by introducing a dynamic, matrix-based method for calculating transformities. This approach incorporates feedback, internal cycling, and flow structure, allowing unit emergy values to reflect system behavior and remain additive across components. The resulting framework—Biosphere Dynamic Empower (BDE)—captures the realized renewable empower structured by biospheric organization. By linking thermodynamic principles with systems modeling, this method advances emergy analysis and provides new tools for evaluating sustainability and energy dynamics in complex ecological systems.

Introduction

Open systems thermodynamics (OST) extends classical thermodynamics to systems that exchange energy and matter with their environment—systems like ecosystems and economies that operate far from equilibrium. Few have contributed more to this field than Howard T. Odum, whose work reframed thermodynamics as a principle of organization, not just constraint.

Odum's development of emergy, the Maximum EmPower Principle, energy systems language, and hierarchical self-organization laid the groundwork for understanding ecological and economic systems as energy-transforming networks. His symbolic diagrams—what he called “energy systems language”—enabled both visualization and simulation of complex systems. With emergy, he introduced a common measure for comparing energy, material, and information flows based on the solar energy required to generate them.

More than a set of methods, Odum offered a way of thinking—one that linked thermodynamics, energy quality, and system function across hierarchical scales—by always asking what role a component plays in sustaining the larger system. His approach was resolutely top-down: to understand any part, he would first examine the system that contains and drives it. If something persisted, Odum assumed it served a function—reinforcing energy flow, stabilizing feedback, or supporting higher-level organization.

In this paper, we first review Odum's contributions to OST and the key ideas that emerged from his systems ecology framework. We then extend this legacy by developing a dynamic, matrix-based method for calculating transformities, inspired by work Odum began late in his career with Dennis Collins. This new approach preserves Odum's emphasis on energy quality and feedback, while offering greater analytical consistency and fidelity to system behavior. The resulting

framework, which we term Biosphere Dynamic Empower (BDE), provides a scalable and additive method for evaluating renewable empower in complex systems.

Classical Thermodynamics and the shift to Open Systems

The historical development of thermodynamics reflects an expanding recognition of complexity, openness, and systemic interaction. As outlined in **Table 1**, the field progressed from the classical foundations established in the 19th century to more integrative formulations in the 20th and 21st centuries. The early work of Sadi Carnot, Rudolf Clausius, William Thomson (Lord Kelvin), and Willard Gibbs defined classical thermodynamics, focusing on closed or isolated systems and idealized processes such as heat engines and chemical equilibria. Carnot's 1824 *Reflections on the Motive Power of Fire* laid the groundwork for later formulations of thermodynamic efficiency, while Clausius introduced the concept of entropy and the second law of thermodynamics, describing the inevitable dissipation of useful energy in isolated systems. Lord Kelvin advanced the concept of absolute temperature, and Gibbs introduced free energy and chemical potential, offering powerful tools for understanding energy transformations in chemical and physical systems.

The transition toward open systems thermodynamics (OST) began in the early 20th century, driven by efforts to explain biological, ecological, and nonequilibrium phenomena. Alfred Lotka (1922a, 1922b) introduced the idea that natural selection favors systems that maximize energy throughput—an early link between energy flow and evolution. Lars Onsager (1931a, 1931b) formalized reciprocal relations in irreversible processes, laying a thermodynamic foundation for open, nonequilibrium systems. Ludwig von Bertalanffy (1950; 1968) developed general systems theory, emphasizing the importance of organization, interaction, and feedback in open biological systems.

The most significant advances in OST came in the second half of the 20th century. Ilya Prigogine (1967) introduced the concept of dissipative structures, showing that open systems far from equilibrium can self-organize through continuous energy flow and dissipation. Odum, however, offered a decisive shift: whereas Prigogine emphasized dissipation, Odum focused on self-organization driven by energy transformation and hierarchical structure. He extended OST by introducing the concepts of energy quality, emergy, and transformity, developing tools to quantify the organization and function of complex ecological and economic systems.

Complementing Odum's work, several contemporaries expanded the application of thermodynamics and systems theory across disciplines. Kenneth Boulding (1966) and Nicholas Georgescu-Roegen (1971) brought thermodynamic reasoning into economics, emphasizing the limits of growth and the role of entropy in economic systems. James G. Miller (1978) extended systems theory to living organisms and social structures, reinforcing the relevance of open systems thinking. In ecology, Richard Ulanowicz (1986, 1997) developed a parallel framework—ascendency theory—to measure ecosystem organization and flow, while Manfred Eigen (1971) and Erwin Schrödinger (1944) advanced understanding of self-organization at molecular and biological levels. Collectively, their work broadened the relevance of thermodynamic and systems-based reasoning, reinforcing its applicability to ecological, biological, and socioeconomic domains.

H.T. Odum's Contributions to Open Systems Thermodynamics

Howard T. Odum made foundational contributions to open systems thermodynamics (OST) by extending thermodynamic principles beyond physics and chemistry into the realms of ecology, economics, and complex systems. His work reframed ecosystems and economies as energy-processing networks that are open to continuous flows of energy, materials, and information from their environment. Through decades of empirical research and conceptual innovation, Odum provided tools for understanding how such systems self-organize, persist, and evolve. We have organized his key contributions into five interrelated themes that illustrate the depth and breadth of his impact:

1. Energy Flow in Ecosystems
2. Energy Systems Language and Diagrams
3. Self-Organization for Maximum EmPower
4. Emergy and Transformity
5. Hierarchical Organization and System Feedbacks

1. Energy Flow in Ecosystems

Odum's earliest and most enduring contribution to open systems thermodynamics was his focus on energy flow as the fundamental organizing principle of ecosystems. Although the formal term "ecosystem" is often attributed to Arthur Tansley, the conceptual foundations of ecosystem energetics were laid earlier by Vladimir Vernadsky (1922)¹, whose work on the biosphere emphasized the planetary-scale movement of energy and matter. Vernadsky's ideas deeply influenced G. Evelyn Hutchinson, who in turn mentored Raymond Lindeman² and H. T. Odum. Lindeman's (1942) classic paper on trophic-dynamic systems marked a turning point in ecosystem theory, proposing that ecological communities could be understood as hierarchies of energy transfer. Odum, as a student of Hutchinson, would take this insight further than anyone before him.

While others had theorized about energy in ecosystems, Odum was the first to systematically and quantitatively measure energy flows in ecological systems (Figure 1), combining Lindeman's trophic-dynamic theory with detailed field measurements. Through this empirical rigor, he demonstrated that ecosystems could be treated as open thermodynamic systems, requiring a continuous input of high-quality energy—primarily solar radiation—to maintain their structure and function. His models traced energy from primary producers through consumers and decomposers, accounting for losses at each stage and showing how energy ultimately dissipated as heat in accordance with the second law of thermodynamics.

Odum's early studies of springs, estuaries, and tropical forests laid the groundwork for what would become systems ecology, an approach that integrates energy flow, matter cycling, and feedback

¹ Although there is no record of direct correspondence between Hutchinson and Vernadsky, Hutchinson cited Vernadsky's work and was influenced by his concept of the biosphere as a system governed by the cycling of energy and materials. This intellectual connection is noted by Polunin and Grinevald, who identify Hutchinson as one of the early Western ecologists to engage with Vernadsky's biospheric perspective. See: Polunin & Grinevald (1986).

² Lindeman was not a formal student of G. Evelyn Hutchinson but shared a draft of his manuscript with him. Hutchinson, recognizing its significance, advocated for its publication after initial rejection by *Ecology*. See Hagen (1992).

control. His diagrams and models illustrated how ecosystems process energy—not in isolation, but as dynamic networks responding to external inputs and internal constraints.

Later in his career, Odum extended these principles beyond natural ecosystems to include human and economic systems, showing that the same thermodynamic laws governed cities, industries, and nations. Unlike classical thermodynamics, which focused on heat engines and physical efficiency, Odum's view emphasized organization and transformation—how energy supports structure, complexity, and feedback control. This insight revealed that energy flows do not just sustain ecosystems, but also shape social and economic hierarchies, linking thermodynamics with the organization of civilization itself.

2. Energy Systems Language and Diagrams

To model the complexity of open systems such as ecosystems, economies, and global cycles, H.T. Odum developed the energy systems language—a symbolic notation system designed to represent the storage, flow, transformation, and feedback of energy within a system. This language provided a visual grammar for thermodynamics, enabling researchers to diagram the energetics of real-world systems in a consistent and comprehensible way. It was not merely illustrative, but analytical, translating system structure into equations that could be simulated and tested.

Odum's diagrams, often referred to as energy circuit diagrams, generalized concepts from electronic circuits to ecological and economic systems. Just as resistors, capacitors, and batteries regulate current in electrical networks, Odum's symbols depicted pathways of energy, storage reservoirs, feedback loops, and control mechanisms. These diagrams helped researchers see how external energy inputs drive system processes, maintain structure, and enable self-organization. As Odum explained,

“Because the existing symbolic and mathematical languages were inadequate to represent the thermodynamics of real ecosystems, we invented the energy systems language as a generalization of electronic circuits”

(Odum, 1995, Chapter 37, Maximum Power).

This invention enabled a new class of *macroscopic minimodels*—models that were broad in scope but simplified enough to be intelligible and tractable. These were not merely theoretical constructs, but dynamic tools used to test hypotheses about system function, organization, and behavior over time.

For Odum, simulation was indispensable to understanding. It provided a means to test hypotheses, explore system dynamics, and observe behaviors that would otherwise unfold too slowly or involve too many variables to interpret directly. Through simulation, the principles of thermodynamics became operational—not just abstract laws but testable drivers of system organization, resilience, and evolution. In essence, the energy systems language brought structure to thermodynamic insight, making the invisible currents of energy visible, traceable, and ultimately knowable in dynamic systems.

3. Self-Organization and the Maximum EmPower Principle

Among Odum's most profound contributions to open systems thermodynamics is the Maximum EmPower Principle³, which posits that systems evolve toward configurations that maximize their power intake, transformation, and use. According to this principle, self-organizing systems optimize their structure not for efficiency alone, but for maximum useful energy flow under prevailing environmental constraints. Odum framed this as a fundamental law of system behavior—one that guides the evolution and persistence of ecological, economic, and technological systems alike. He often referred to it as a thermodynamic analog to Darwinian fitness.

This view extends classical thermodynamics by moving beyond equilibrium and dissipation. While Ilya Prigogine described far-from-equilibrium systems as dissipative structures, Odum found the term problematic. "Dissipation" emphasizes energy loss, overshadowing the constructive, self-organizing nature of complex systems. In contrast, Odum's use of the term "self-organizing systems" better highlights the emergent order and feedback mechanisms that arise from energy flow. It shifts attention from decay to creativity, aligning more intuitively with the observable behavior of ecosystems, economies, and living systems.

Odum's models showed that systems do not merely persist—they develop feedback loops, recycling pathways, and control structures that enhance their ability to capture and utilize energy. As illustrated in [Figure 2](#), self-organizing systems evolve structures that reinforce their energy-gathering capabilities, making them more resilient and adaptive to changing conditions. The flow of energy, matter, and information becomes a driver of increasing complexity—not merely a fuel source.

The Maximum EmPower Principle, as Odum refined it, is often oversimplified as a trade-off between efficiency and throughput. While this early formulation—developed with Richard Pinkerton—has intuitive appeal, it falls short of conveying the full dynamism of the principle. As Alfred Lotka originally articulated:

"The Maximum Power Principle reveals itself as capable of yielding information which the first and second laws of thermodynamics are not competent to furnish. The two fundamental laws of thermodynamics are, of course, insufficient to determine the course of events in a physical system. They tell us that certain things cannot happen, but they do not tell us what does happen."

Odum's insight provided a thermodynamic basis for evolution, adaptation, and system design, connecting energetics to self-organization in a way that classical and even Prigoginian thermodynamics could not. Through the lens of maximum empower, ecosystems and human systems are no longer seen as passive energy dissipators, but as active, evolving agents shaped by and shaping their energy environment.

³ The Maximum Power Principle was originally formulated by Alfred J. Lotka in the early 20th century, particularly in his 1922 papers on natural selection and energy flow. Odum recognized the principle's fundamental importance and reintroduced it in the context of systems ecology, referring to it early on as "Lotka's Maximum Power Principle." Odum expanded its application beyond biology to ecosystems, economies, and engineered systems, establishing it as a foundational law of open systems thermodynamics.

4. Energy Quality, Emergy, and Transformity

Perhaps the most innovative—and often least understood—of Odum's contributions to open systems thermodynamics is his concept of energy quality, which he formalized through the constructs of emergy and transformity. Recognizing that not all energy is equal in its ability to do work, Odum defined emergy as the total available energy of one form—typically solar—required directly and indirectly to generate a product, service, or flow within a system. This formulation provides a way to express all energy inputs on a common basis, allowing meaningful comparisons across different energy forms and systems.

Emergy offers a comprehensive accounting tool for evaluating ecological and economic processes in terms of their true energy requirements, making it especially valuable for sustainability assessments and resource-use evaluations. It bridges thermodynamic principles with real-world system functioning by tracing the cumulative energy transformations that support any given output.

At the core of this framework is transformity—the emergy per unit of output energy. Transformity reflects energy quality and serves as a metric for positioning processes and components within an energetic hierarchy. High-transformity outputs—such as human services or digital information—require many transformations of lower-quality energy, reinforcing Odum's view that hierarchy emerges naturally through energy processing and feedback.

By introducing emergy and transformity, Odum extended systems theory beyond conventional concerns with efficiency or entropy. These concepts reveal the underlying energetics that govern how systems organize, persist, and evolve—by capturing, transforming, and concentrating energy across levels of quality. Emergy provides a common metric for tracing energy memory through ecological and human networks, while transformity clarifies the hierarchy of energy use. Together, they expose the energetic logic behind feedback, storage, and self-organization in complex systems.

5. Hierarchical Organization and System Feedbacks

A central tenet of Odum's contributions to open systems thermodynamics is the idea that natural and human systems are hierarchically organized. In this view, ecosystems are not just collections of interacting parts but nested structures in which components at different levels—organisms, populations, communities, and biomes—function on distinct spatial and temporal scales. Each level is governed by feedback mechanisms that regulate the flow of energy and materials, helping the system maintain stability and adapt to change.

Odum's models demonstrated that hierarchical organization emerges naturally from the flow of energy. As energy is transformed by successive components, it moves through a cascade of processes, each operating at different intensities, with distinct delays and control functions. At each level, the quantity of energy decreases due to 2nd Law losses, but its quality increases as it is concentrated into forms capable of more specialized work. Reinforcing feedback loops link these hierarchical levels, coordinating energy use and shaping system behavior across scales.

Odum introduced the concept of energy quality as the key principle structuring the energy hierarchy (see Figure 3). He proposed that systems self-organize in a way that concentrates energy into forms of higher quality—that is, forms capable of doing more work per unit of energy. The concept of transformity—defined as the amount of energy of one type required to generate a unit

of another, higher-quality form—was Odum’s key metric for positioning processes within this energy-based hierarchy.

Transformity provides a quantitative means for understanding the structure of complex systems. Low-transformity inputs like sunlight support vast primary productivity at the base of the ecological hierarchy, while high-transformity outputs—such as human labor, information, or cultural institutions—occupy the upper levels, supported by extensive energy transformations. In this way, transformity bridges energetics and structure, allowing systems to be analyzed not just by their components, but by the roles those components play in organizing and directing the flow of energy.

By placing energy quality at the heart of hierarchy and feedback, Odum offered a thermodynamically grounded explanation for why complex systems evolve the way they do—why structure, control, and scale emerge from energy flow, and why systems succeed or fail depending on their ability to regulate that flow effectively.

Summary of Odum’s Impact on Open Systems Thermodynamics

Odum’s contributions have been foundational in applying open systems thermodynamics to the study of ecological and economic systems as energy-dependent, self-organizing networks. His development of concepts such as emergy, the Maximum EmPower Principle, and hierarchical organization provided a unified framework for measuring and interpreting energy flows across diverse systems. By quantifying energy quality and emphasizing feedback and transformation, Odum deepened our understanding of the functional interdependence between nature and human economies. His work offered a thermodynamically grounded way to evaluate sustainability—not as a static condition, but as a dynamic, energy-based process.

To honor H.T. Odum’s legacy, we restate the classical laws of thermodynamics—originally formulated for closed systems—in terms more appropriate for the behavior of open systems, incorporating two additional principles Odum identified as essential to understanding self-organizing processes in nature and society.

Classical and OST-Extended Laws of Thermodynamics

Classical	OST-Extended
<i>1st Law</i>	
Energy cannot be created or destroyed, only transformed or transferred.	In open systems, energy and material flow across boundaries, conserving total energy while transforming form and enhancing availability.
<i>2nd Law</i>	

In any energy transfer or transformation, entropy increases, and energy becomes less available for work.

Open systems sustain localized order by self-organizing inflows of energy, material, and information and exporting entropy to their surroundings

3rd Law

At absolute zero (0 K), the entropy of a perfect crystalline structure is exactly zero.

As energy gradients diminish, an open system's capacity to generate order, sustain function, and enable transformation declines. At zero availability, self-organization ceases.

4th Law (proposed)

Open systems evolve to maximize energy capture and use by enhancing their capacity to sustain organization, adapt, and perform work.

5th Law (proposed)

Open systems self-organize into hierarchies, with each level transforming and concentrating energy to sustain, regulate, and expand system organization.

Odum's systems framework remains one of the most comprehensive efforts to unify thermodynamics, ecology, and human systems. Yet in his final years, he recognized that existing methods for computing transformities—especially in systems with multiple, interacting outputs—lacked internal consistency and scalability. With Dennis Collins (Collins & Odum, 2000; Odum & Collins, 2002), he began exploring the use of matrix algebra to calculate transformities, laying the groundwork for what would become a dynamic systems approach. Although this work was still in its early stages at the time of his death, it signaled a shift toward a more formal and scalable method of tracing empower in complex systems.

In the following section, we build upon Odum's interest in using matrix methods to calculate transformities—as a promising new direction for emergy analysis. Drawing from these foundations, we develop a dynamic framework for evaluating empower at the biospheric scale, one that retains Odum's core principles while offering new clarity and consistency in accounting for energy flows, feedbacks, and hierarchy in open systems.

Biosphere Dynamic Empower (BDE) Framework

In Environmental Accounting (1996), H.T. Odum introduced the global, top-down approach to evaluate the renewable sources of emergy driving biosphere processes as a static accounting method and refined it in Folio #1: Introduction and Global Budget (Odum, 2000). Assuming the

biosphere operates in steady state and without internal recycling, he calculated Unit Energy Values (UEVs)⁴ by dividing the total solar equivalent energy of the Earth's surface—what we now call the Geobiosphere Emergy Baseline (GEB)—by the global annual exergy of each secondary or tertiary renewable source⁵:

$$UEV_{\text{source}} = \frac{GEB \text{ (sej/yr)}}{\text{Renewable Source Energy (J/yr)}}$$

This approach treats each renewable source (e.g., wind, rain, waves, etc.) as a global end-product of biospheric self-organization, co-products of the GEB, and yields what can be considered an upper-bound UEV—reflecting the full emergy investment of the planet into that flow. An important consideration is how these static UEVs are used in land-based evaluations. Since each secondary source is computed using the full GEB as its numerator, summing multiple renewables (e.g., rain, wind, river flow) would result in double counting. Therefore, the correct approach is to identify and use only the largest renewable emergy input among them. This ensures that based on static rules, emergy is not double counted.

Dynamic Matrix Approach

To extend Odum's legacy and move beyond static emergy accounting, we implement a dynamic matrix-based method for calculating transformities of the secondary and tertiary renewable flows of the biosphere. This approach begins by constructing a system of energy flows among biospheric compartments that represent renewable processes. External emergy inputs—sourced from the Geobiosphere Empower Base (GEB)—are assigned to each compartment, and unit emergy values (UEVs) are derived by solving a set of simultaneous equations using matrix inversion. Unlike static models, this method explicitly incorporates feedback, internal cycling, and emergy reinforcement along interconnected pathways.

To highlight the methodological divergence from Odum's original approach, **Table 2** compares key features of the static and dynamic methods. In the static method, emergy inputs are allocated independently for each output by dividing the total GEB by the quantity of that product, without regard for internal system structure or interdependence. In contrast, the dynamic method allocates inputs according to the actual distribution of energy flows within the system. Feedback loops, recycling pathways, and flow topology shape the outcome, yielding UEVs that more accurately reflect the role and function of each system component.

The biospheric energy systems diagram in **Figure 4** illustrates the compartments and external driving inputs. These relationships are formalized in a flow matrix (**Table 3**), where the transfer of emergy among components is tabulated. For analysis, the matrix is transposed—so that columns

⁴ Unit Emergy Value (UEV) is a general term encompassing several specific emergy-based metrics: transformity (solar emergy per unit energy, sej/J), specific emergy (solar emergy per unit mass, sej/g), and emprice (solar emergy per unit monetary cost, sej/\$).

⁵ In recent years the renewable emergy sources have been organized into three main types, primary sources (solar, tidal, and geothermal), secondary sources (wind, rain, ocean currents), and tertiary sources (waves, river chemical potential, river geopotential)

correspond to the inputs needed to generate each output—and multiplied by -1 to match matrix bookkeeping conventions. The resulting system is solved as:

$$(-A^T) \cdot x = b$$

yielding a vector of D-Transformities—emergy per unit energy—for each biospheric output (the last column in [Table 3](#)).

The transformities and total empower values derived using this approach are shown in [Table 4](#). One of the critical advantages of the dynamic method is that it yields additive UEVs. Because each compartment receives only its proportion of the total GEB, and outputs are evaluated independently with respect to that distribution, the resulting values can be summed without double-counting. This makes D-Transformities particularly suitable for system-level emergy accounting, where additive properties are essential.

Importantly, the total renewable empower calculated across all flows in the dynamic system exceeds the raw GEB input. This emergy amplification arises from feedback and internal cycling, which reinforce energy use across trophic and spatial hierarchies. We define this cumulative, systems-based quantity as the *Biosphere Dynamic Empower (BDE)*—a measure of the total renewable empower structured and sustained by the biosphere’s internal organization. Unlike the GEB, which represents boundary inputs to the Earth system, the BDE reflects the realized empower available to support ecosystem function, evolution, and adaptation.

Static and Dynamic Compared

[Table 5](#) lists both D-transformities and static transformities for key renewable flows. In most cases, D-transformities are lower than their static counterparts—not because less emergy is assigned, but because the emergy input to each flow is proportionally partitioned rather than fully allocated as in the static method. Static transformities often reflect the entire GEB assigned to a single output, inflating the unit emergy value. By contrast, the dynamic method distributes the GEB among all flows and includes recycled emergy through feedback pathways. While this results in lower transformities, it allows renewable flows to be summed without redundancy, often yielding a higher total emergy contribution than under the Max-R⁶ approach.

Redefining Renewable Empower in a Dynamic Biosphere

In traditional emergy accounting, the Geobiosphere Emergy Baseline (GEB)—the combined solar, tidal, and geothermal exergy—served as the defining measure of renewable input to Earth’s systems. This made sense under a static framework, where emergy was assumed to enter the system once and drive processes without internal reinforcement.

Under the dynamic approach, however, secondary and tertiary renewable flows—rain, wind, waves, river energy—represent real, regenerating empower that circulates through the biosphere, enabling work again and again. These flows emerge not directly from boundary inputs, but from the internal organization, feedback, and structure of the biosphere. As such, they represent

⁶ Max-R Rule: A convention used in emergy accounting to avoid double-counting when evaluating areal systems. The total renewable input is taken as the greater of either the Geobiosphere Empower Base (GEB) or the largest individual secondary or tertiary renewable flow (e.g., rain, wind, etc). This rule assumes that each derived renewable flow reflects the full GEB, and therefore these flows cannot be summed without redundancy.

the realized renewable work potential of the system, which we define as the *Biosphere Dynamic Empower (BDE)*.

When comparing the total renewable emergy derived from dynamic modeling (Table 4; 40.0×10^{24} sej/yr) to the GEB alone (12.0×10^{24} sej/yr), we observe more than a threefold increase in effective renewable empower. This amplification is not an artifact—it reflects the biosphere's ability to multiply and redirect incoming empower through cycles, storages, and feedback loops. The BDE is thus an emergent property of biospheric structure and memory.

This redefinition of renewable empower has direct implications for emergy assessments of land and ocean systems. Whereas earlier methods focused only on boundary inputs or selected dominant flows, the dynamic framework enables the summation of multiple biosphere-driven inputs. This not only enhances the accuracy of emergy analysis but also captures the depth, feedback, and complexity of biospheric function.

Extending Odum's Vision—A Dynamic, Quality-Based Thermodynamics of Open Systems

Throughout this paper, we have traced Howard T. Odum's foundational contributions to open systems thermodynamics, including his pioneering concepts of energy quality, emergy, transformity, and the hierarchical organization of self-organizing systems. By applying thermodynamic principles to ecological and economic processes, Odum developed a systems framework for understanding how energy flows, feedbacks, and storages drive the organization and persistence of complex systems. Building on this foundation, we have extended his vision by introducing a dynamic, matrix-based method for calculating transformities of the renewable emergy flows of the biosphere—a method that explicitly incorporates internal energy flows, feedback structures, and emergy cycling. The resulting Biosphere Dynamic Empower (BDE) framework offers a refined approach to quantifying renewable empower in a way that reflects system behavior and structure.

This dynamic approach builds directly on Odum's foundational principles while addressing limitations inherent in static emergy accounting. By deriving additive, system-specific transformities from actual flow structures, the BDE framework provides a more internally consistent method for evaluating energy use and organization in complex systems. It offers a way to incorporate feedback, cycling, and system configuration into emergy-based modeling, improving alignment with the thermodynamic behavior of real systems. As interest grows in applying thermodynamic insights to sustainability, ecological resilience, and resource planning, the BDE approach contributes a practical step toward more dynamic, structure-sensitive evaluations. In doing so, it maintains continuity with Odum's systems thinking while extending it with tools suited to contemporary modeling challenges.

Literature Cited

- Boulding, K. E. (1966). *The economics of the coming spaceship Earth*. In H. Jarrett (Ed.), *Environmental quality in a growing economy* (pp. 3–14). Baltimore, MD: Johns Hopkins University Press.
- Collins, D. and H.T. Odum. 2000. Calculating transformities with an eigenvalue method. pp.

- 265-280 in *Emergy Synthesis: Theory and Applications of the Emergy Methodology*, Proceedings of the International Workshop on Emergy and Energy Quality, Gainesville, FL, Sept. 1999, ed. by M.T. Brown. Center for Environmental Policy, Univ. of Florida, Gainesville, 328 pp.
- Davies, J. H., & Davies, D. R. (2010). Earth's surface heat flux. *Solid earth*, 1(1), 5-24.
- Davies, J. H., & Davies, D. R. (2010). Earth's surface heat flux. *Solid earth*, 1(1), 5-24.
- Egbert & Ray (2000), Significant dissipation of tidal energy in the deep ocean inferred from satellite altimeter data. *Nature*. 405(6788), 775-778.
- Egbert & Ray (2000), Significant dissipation of tidal energy in the deep ocean inferred from satellite altimeter data. *Nature*. 405(6788), 775-778.
- Eigen, M. (1971). Self-organization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58(10), 465–523.
<https://doi.org/10.1007/BF00623322>
- Georgescu-Roegen, N. (1971). *The entropy law and the economic process*. Cambridge, MA: Harvard University Press.
- Hagen, J.B. (1992). *An Entangled Bank: The Origins of Ecosystem Ecology*, Rutgers University Press, pp. 70–75.
- Hofmeister, A.M. and Criss, R.E., Earth's heat flux revised and linked to chemistry, *Tectonophysics*, Volume 395, Issues 3-4, 20 January 2005, Pages 159-177.
- Hofmeister, A.M. and Criss, R.E., Earth's heat flux revised and linked to chemistry, *Tectonophysics*, Volume 395, Issues 3-4, 20 January 2005, Pages 159-177.
- Kiehl, J. T., & Trenberth, K. E. (1997). Earth's annual global mean energy budget. *Bulletin of the American meteorological society*, 78(2), 197-208.
- Kiehl, J. T., & Trenberth, K. E. (1997). Earth's annual global mean energy budget. *Bulletin of the American meteorological society*, 78(2), 197-208.
- Lindeman, R. L. (1942). The trophic-dynamic aspect of ecology. *Ecology*, 23(4), 399–417.
<https://doi.org/10.2307/1930126>
- Lotka, A. J., 1922a. "Contribution to the energetics of evolution". *Proc Natl Acad Sci USA*, 8: pp. 147–51.
- Lotka, A. J., 1922b. "Natural selection as a physical principle". *Proc Natl Acad Sci USA*, 8, pp. 151–54.
- Miller, J. G. (1978). *Living systems*. New York: McGraw-Hill.
- Munk, W., & Wunsch, C. (1998). Abyssal recipes II: Energetics of tidal and wind mixing. *Deep Sea Research Part I: Oceanographic Research Papers*, 45(12), 1977-2010.)
- Munk, W., & Wunsch, C. (1998). Abyssal recipes II: Energetics of tidal and wind mixing. *Deep Sea Research Part I: Oceanographic Research Papers*, 45(12), 1977-2010.)
- Odum, H. T. and R. C. Pinkerton. 1955. Time's speed regulator: the optimum efficiency for maximum output in physical and biological systems. *Am. Sci.* 43:331-343.

- Odum, H.T. 2000. Handbook of Emergy Evaluation: A Compendium of Data for Emergy Computation Issued in a Series of Folios. Folio #2 - Emergy of Global Processes. Center for Environmental Policy, Environmental Engineering Sciences, Univ. of Florida, Gainesville, 30 pp.
- Odum, H.T. (1971). Environment, Power, and Society. Wiley-Interscience.
- Odum, H.T. (1988). Self-Organization, Transformity, and Information. *Science*, 242(4882), 1132-1139.
- Odum, H.T. (1996). Environmental Accounting: Emergy and Environmental Decision Making. John Wiley & Sons, New York. 370p.
- Odum, H.T. 1956. Primary production in flowing waters. *Limnol. and Oceanogr.* 1:102-117.
- Odum, H.T. and D. Collins 2002. More Transformities from Ecosystem Energy Webs with the Eigenvalue Method. Chapter in Emergy Synthesis: Theory and Applications of the Emergy Methodology. Proceedings of the Sept. 2001 Second Biennial Emergy Analysis Research Conference, Gainesville, FL, ed. by M.T. Brown. Center for Environmental Policy, Univ. of Florida, Gainesville.
- Odum, H.T. and Odum, E.C., 1976. Energy basis for man and nature. McGraw-Hill, New York. 337p.
- Odum, H.T., 1983. Systems Ecology; an introduction. John Wiley and Sons. New York. 644p.
- Odum, H.T.(2007). Environment, Power, and Society for the 21st Century. Columbia University Press, New York. 418p.
- Onsager, L. (1931). Reciprocal relations in irreversible processes. I. *Physical Review*, 37(4), 405–426.
- Onsager, L. (1931). Reciprocal relations in irreversible processes. II. *Physical Review*, 38(12), 2265–2279.
- Pollack, H. N., Hurter, S. J., & Johnson, J. R. (1993). Heat flow from the Earth's interior: analysis of the global data set. *Reviews of Geophysics*, 31(3), 267-280.
- Pollack, H. N., Hurter, S. J., & Johnson, J. R. (1993). Heat flow from the Earth's interior: analysis of the global data set. *Reviews of Geophysics*, 31(3), 267-280.
- Polunin, N., & Grinevald, J. (1986). Vernadsky and biospherical ecology. *Environmental Conservation*, 13(3), 233–240. <https://doi.org/10.1017/S0376892900035200>
- Prigogine, I. (1967). Introduction to thermodynamics of irreversible processes (3rd ed.). New York: Interscience Publishers.
- Ray & Egbert (2012), Tidal energy dissipation in the solid Earth. *J. Geophys. Res.*
- Ray, R. D., & Egbert, G. D. (2012). Fortnightly Earth rotation, ocean tides and mantle anelasticity. *Geophysical Journal International*, 189(1), 400-413.
- Schrödinger, E. (1944). *What is Life? The physical aspect of the living cell*. Cambridge: Cambridge University Press.
- Stein, C. A., & Stein, S. (1992). A model for the global variation in oceanic depth and heat flow with lithospheric age. *Nature*, 359(6391), 123-129.

- Stein, C. A., & Stein, S. (1992). A model for the global variation in oceanic depth and heat flow with lithospheric age. *Nature*, 359(6391), 123-129.
- Trenberth, K. E., Fasullo, J. T., & Kiehl, J. (2009). Earth's global energy budget. *Bulletin of the American Meteorological Society*, 90(3), 311-324.
- Trenberth, K. E., Fasullo, J. T., & Kiehl, J. (2009). Earth's global energy budget. *Bulletin of the American Meteorological Society*, 90(3), 311-324.
- Ulanowicz, R. E. (1986). *Growth and development: Ecosystems phenomenology*. New York: Springer-Verlag.
- Ulanowicz, R. E. (1997). *Ecology, the ascendent perspective*. New York: Columbia University Press.
- von Bertalanffy, L. (1950). An outline of general system theory. *British Journal for the Philosophy of Science*, 1(2), 134-165.
- von Bertalanffy, L. (1968). *General system theory: Foundations, development, applications*. New York: George Braziller.
- Wild, M., Folini, D., Schär, C., Loeb, N., Dutton, E. G., & König-Langlo, G. (2013). The global energy balance from a surface perspective. *Climate dynamics*, 40, 3107-3134.
- Wild, M., Folini, D., Schär, C., Loeb, N., Dutton, E. G., & König-Langlo, G. (2013). The global energy balance from a surface perspective. *Climate dynamics*, 40, 3107-3134.
- Wunsch (2000), Moon, tides, and climate. *Nature*. 405(6788), 743-744.

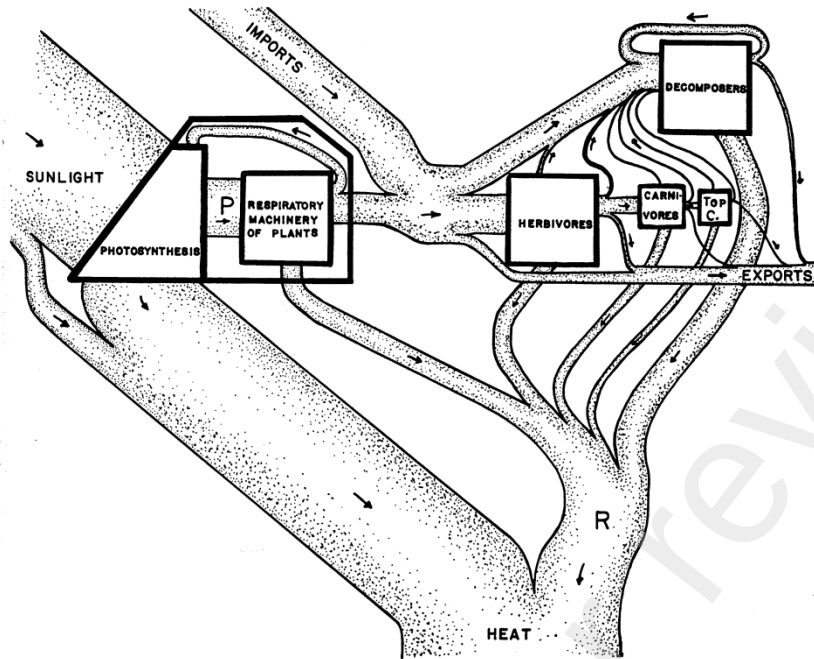


Figure 1. Energy flow in a “generic ecosystem.” The width of line is proportional to the quantity of energy. Arrows indicate direction of flow. P = gross primary production, R = respiration
(Odum, H.T. 1956)

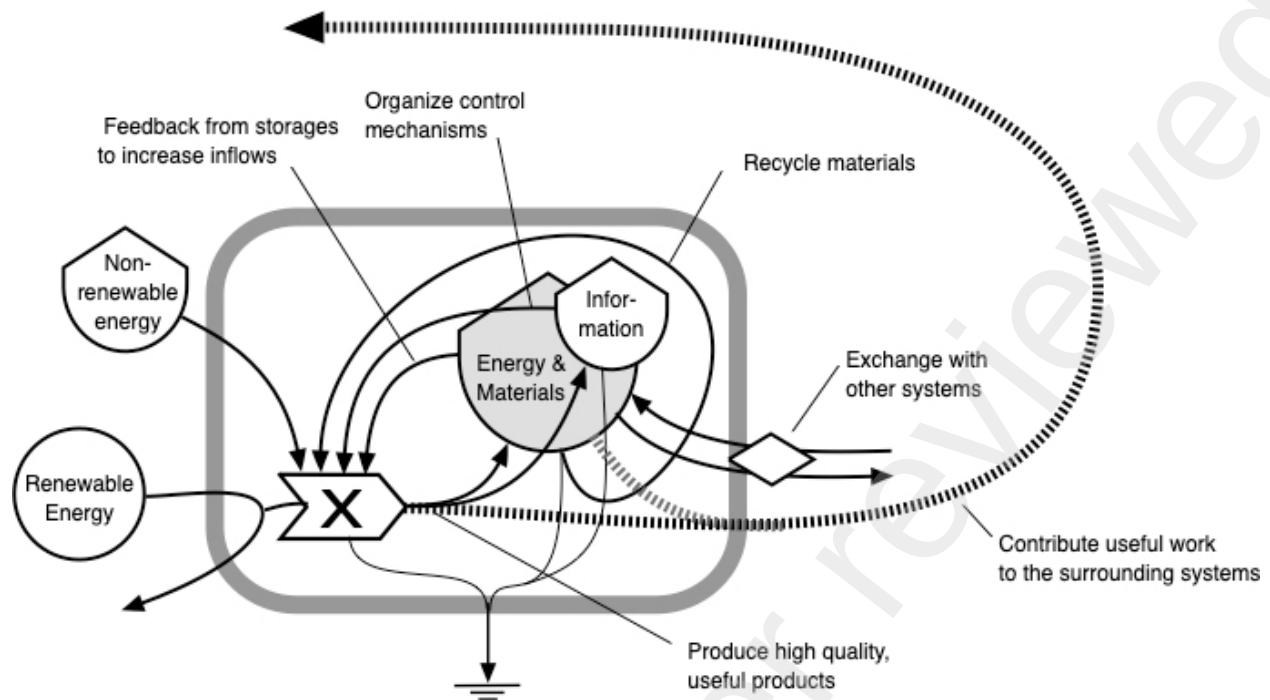


Figure 2. Odum expanded on Lotka's Maximum Power Principle by proposing six corollaries that describe how systems maximize power. (1) Develop storages of energy, materials, and information—emphasizing that systems are not passive conduits but producers. (2) Establish feedbacks to reinforce and increase inflows. (3) Recycle materials to conserve resources internally. (4) Organize control flows, highlighting the role of information in system regulation. (5) Trade with other systems to acquire scarce resources. (6) Contribute useful work to the next larger system, creating reinforcing reward loops across scales. (Odum and Odum, 1976)

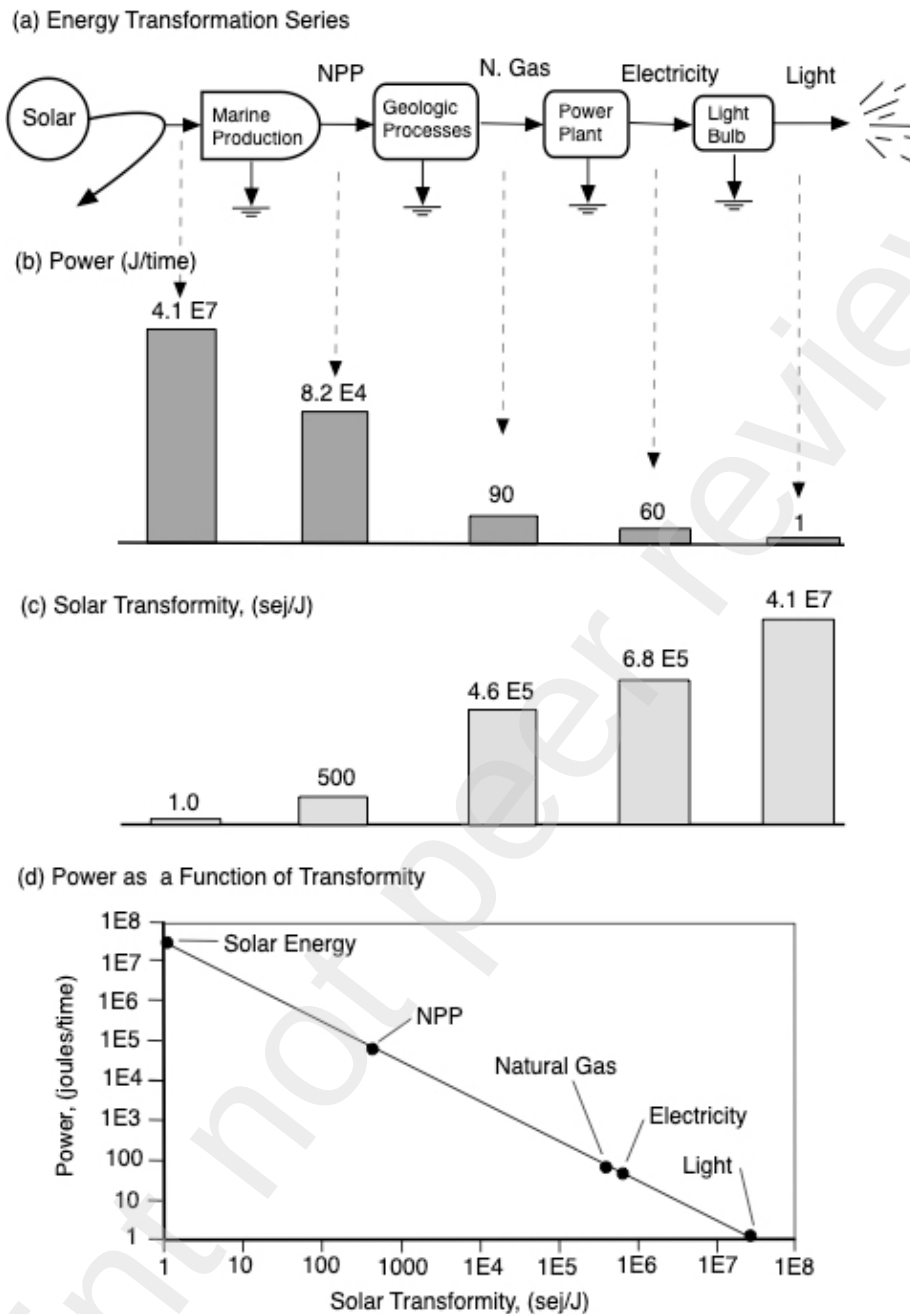


Figure 3. Energy flows that drive systems naturally generate hierarchical organization, with quality differences in energy emerging from second-law losses. Transformity reflects hierarchical position. (a) Energy chain from marine production to an electric light; (b) Power flows in joules per unit time; (c) Solar transformities, calculated by dividing solar input by the energy at each stage, (d) graph of power as a function of transformity. (After Odum, 2007)

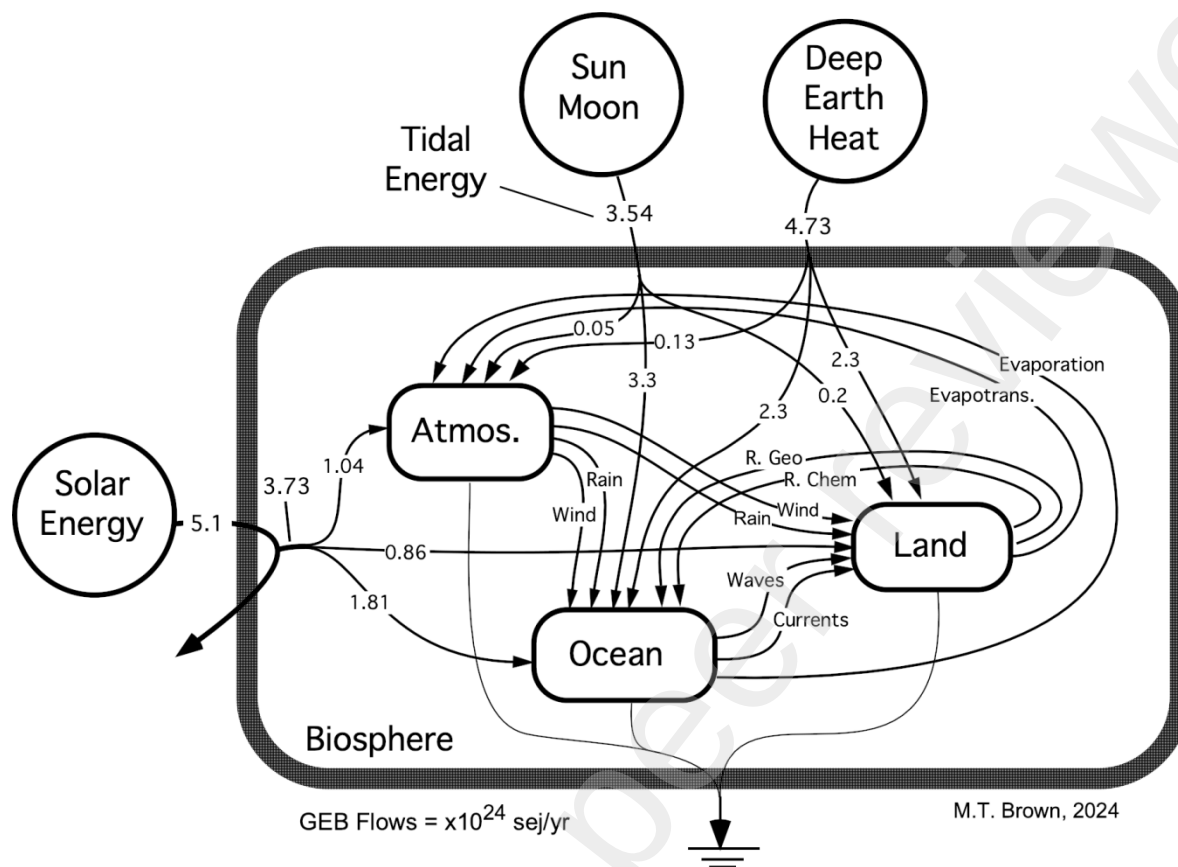


Figure 4. Earth biosphere, a hierarchical web of components connected by flows of available energy and materials that build potential energy and circulate materials. The GEB is split between the atmosphere, ocean and land, based on the following data:

Tidal Emery Dissipation			
Compartment	% Absorbed	Emery (seJ/y)	References
Atmosphere	1.3%	4.53E+22	Munk and Wunsch, 1998); Egbert &
Land	5.5%	1.95E+23	Ray (2000); Wunsch (2000); Ray &
Ocean	93.2%	3.30E+24	Egbert (2012)
	100%	3.54E+24	
Geothermal emery absorption			
Atmosphere	2.7%	1.30E+23	Pollack, Hurter & Johnson (1993);
Land	48.6%	2.30E+24	(Hoffmeister and Criss, 2005) ; Davies
Ocean	48.6%	2.30E+24	& Davies (2010); Stein & Stein (1992)
	100%	4.73E+24	
Solar Emery absorption			
Atmosphere	28.1%	1.04E+24	Trenberth et al.;(2009); Wild et al.
Land	23.1%	8.58E+23	(2013); Kiehl & Trenberth (1997)
Ocean	48.7%	1.81E+24	
	100%	3.71E+24	

Table 1. Historical development of thermodynamics

Researcher	Contribution
Sadi Carnot (1824)	Early studies on the efficiency of heat engines laid the groundwork for the laws of thermodynamics, focusing on closed systems
Rudolf Clausius (1850s)	Introduced the concept of entropy and formalized the second law of thermodynamics, emphasizing energy dissipation in isolated systems.
William Thomson (Lord Kelvin) (1850s)	Defined absolute temperature and worked on energy conservation and entropy within classical thermodynamics
Willard Gibbs (1876)	Developed the concept of free energy (Gibbs free energy); Introduced the idea of chemical potential
Alfred Lotka (1922)	Natural selection operates to maximize energy flux through systems
Lars Onsager (1931)	Developed the reciprocal relations in irreversible processes
Ludwig von Bertalanffy (1950)	Developed General Systems Theory, emphasizing the interconnectivity of open systems in biology and other fields.
Ilya Prigogine (1967)	Introduced the concept of dissipative structures, where open systems self-organize under the influence of energy flows
H.T. Odum (1950s-2012)	Pioneered the application of thermodynamic principles to ecosystems and economies. Introduced self-organization for maximum power energy quality, emergy, and transformity

Table 2. Comparison of Odum’s original static emergy evaluation method and the matrix-based dynamic method.

Static Method	Dynamic Method
Global, top-down <i>Uses a single, biosphere-scale perspective</i>	System-specific, bottom-up <i>Derived from internal structure of a defined system</i>
UEVs as end-products of the biosphere <i>Computed by dividing total biosphere emergy by output emergy</i>	UEVs as outcomes of internal system flows <i>Emergy values reflect internal pathways and allocations</i>
No feedbacks or internal cycling <i>Assumes linear, non-recursive emergy transfer</i>	Feedback, cycling, and reinforcements included <i>Captures network effects and recursive flows</i>
Single global GEB applied to all components <i>All outputs share the same total emergy base</i>	Emergy inputs partitioned across system flows <i>Each output receives emergy proportional to its pathway and role</i>

Table 3. From-to-matrix of energy flows and driving emergy used to compute dynamic transformities of biosphere secondary and tertiary renewable emergy

FROM TO	Ocean Wind	Terrestrial Wind	Ocean Rain	Terrestrial Rain	Ocean Evaporation	Ocean Waves	Ocean Currents	Terrestrial Evapotrans.	River Chem. Pot.	River Geo. Pot.	Emergy Vector	UEVs ¹ . (sej/J)
Ocean Wind	1.06E+22	0	0	0	-8.10E+20	0	0	0	0	0	1.06E+23	450
Terrestrial Wind	0	1.20E+22	0	0	0	0	0	-1.90E+20	0	0	7.08E+22	540
Ocean Rain	0	0	1.95E+21	0	-1.13E+21	0	0	0	0	0	7.79E+23	3700
Terrestrial Rain	0	0	0	5.2E+20	-2.10E+20	0	0	-1.6E+20	0	0	2.24E+23	12900
Ocean Evaporation	-3.32E+21	0	-1.36E+21	0	2.14E+21	0	0	0	-8.7E+18	0	5.60E+24	5700
Ocean Waves	-5.40E+21	0	0	0	0	7.57E+19	0	0	0	0	3.74E+23	36800
Ocean Currents	-1.88E+21	0	-5.85E+20	0	0	0	9.47E+19	0	-1.65E+20	0	1.49E+24	70900
Terrestrial Evapotranspiration	0	-1.20E+22	0	-1.78E+20	0	0	0	3.45E+20	0	0	2.85E+24	33900
River Chem. Potential	0	0	0	-1.78E+20	0	0	0	0	1.74E+20	0	6.70E+22	13500
River Geo. Potential	0	0	0	-1.78E+20	0	0	0	0	0	2.87E+20	4.36E+23	9400

1. Rounded to 1 or 2 significant figures

Preprint not peer reviewed

Table 4. Available energy, dynamic transformities, and emergy of secondary and tertiary renewable sources

Component	Energy ¹ (x10 ²¹ J/yr)	D-Transformity ² (sej/J)	Emergy ³ (x10 ²⁴ sej/yr)
Ocean Wind	10.6	450	4.8
Terrestrial Wind	12.50	540	6.8
Ocean Rain	1.95	3,700	7.2
Terrestrial Rain	0.53	12,900	6.9
Shore Waves	0.08	36,800	2.8
Ocean Currents	0.09	70,900	6.7
River Chemical Potential	0.17	13,500	2.3
River Geopotential	0.29	9,400	2.7
Total			40.0

1. Diagonal values in Table 5-5

2. Rounded to 2 significant figures

3. Product of available energy (column 1) and D-Transformity (column 2).

Table 5. Comparison of D-transformities and statically computed transformities of biosphere renewable energies

Renewable Flow	D-Transformity (sej/J)	Static Transformities* (sej/J)
Ocean Wind	450	
Terrestrial Wind	540	520
Ocean Rain	3700	
Terrestrial Rain	12900	22500
Shore Waves	36800	158500
Ocean Currents	70900	126700
River Chemical Potential	13500	68300
River Geopotential	9400	41100

*Static transformities computed using Odum's method by dividing the Geobiosphere Empower Base ($GEB = 12.0 \times 10^{24}$ sej/yr) by the annual available energy of each flow